NETWORK TRAFFIC ANALYSIS USING WIRESHARK

Internship Project Report

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INTRODUCTION

Network Traffic Analysis is the process of observing and understanding the data that travels through a network. Whenever we open a website, watch a YouTube video, or search on Google, data packets are sent and received through the internet. In this project, I used Wireshark software to capture and analyze network traffic. This helped me understand how different protocols work and how devices communicate over a network.

OBJECTIVE

- To understand the basics of network traffic analysis.
- To learn packet capturing using Wireshark.
- To analyze DNS, TCP and TLS protocols.
- To gain practical experience in network monitoring.

TOOLS USED

Wireshark is a packet analyzer used for capturing and analyzing network traffic.

tcpdump is a command-line tool used to capture network packets.

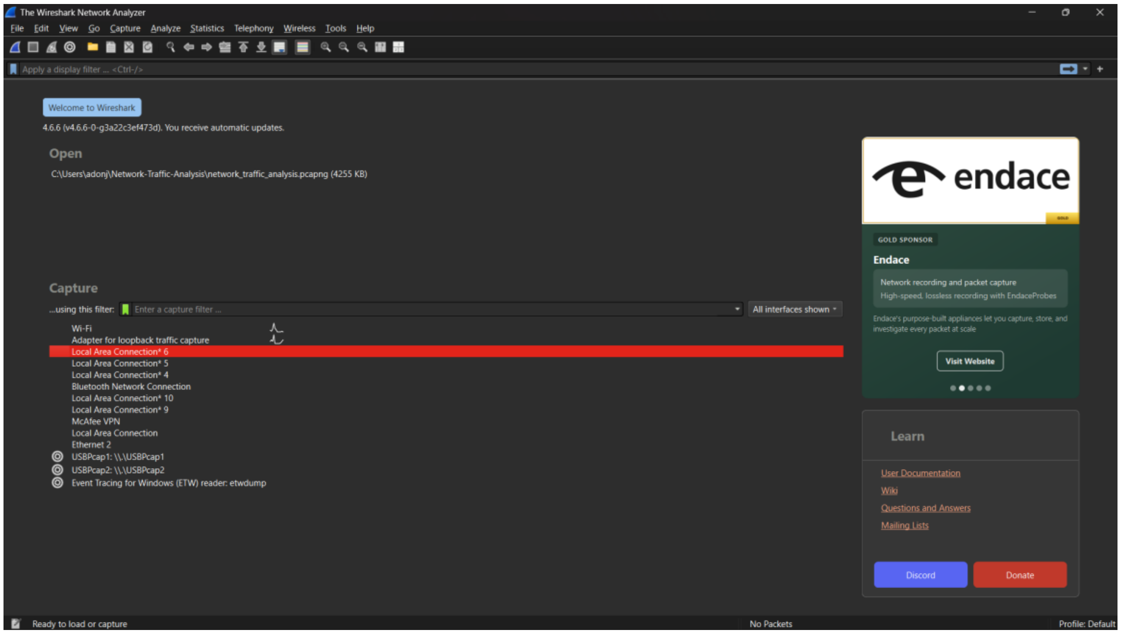
Snort is an intrusion detection tool used for identifying suspicious network activities.

For this project, Wireshark was mainly used.

PROCEDURE

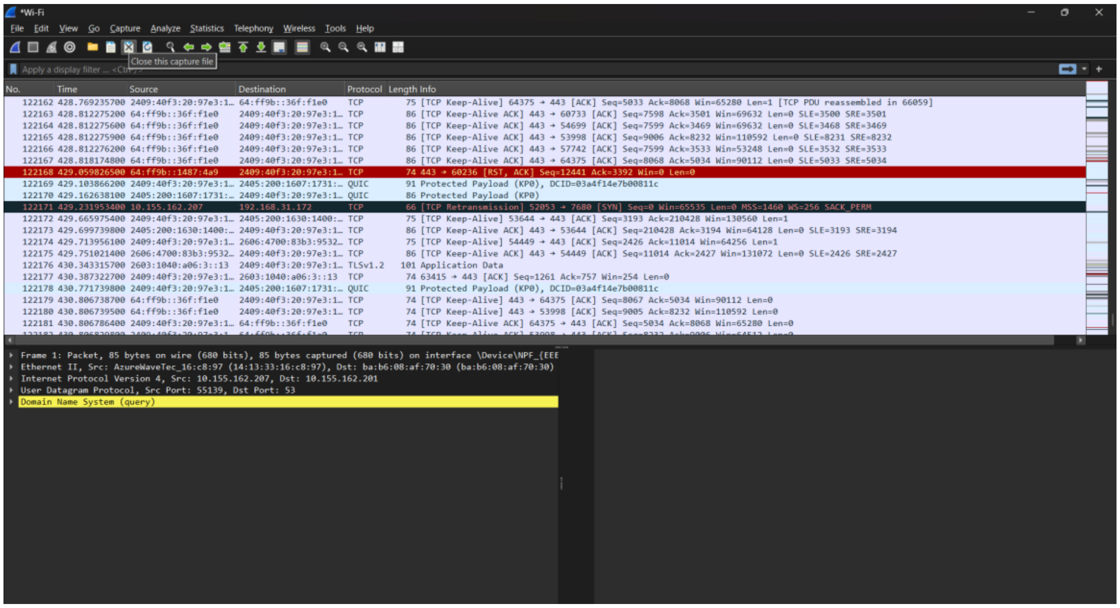
1. Installed Wireshark and opened the application.
2. Selected the Wi-Fi interface and started packet capture.
3. Generated traffic by searching on Google, opening YouTube and visiting websites.
4. Stopped the capture after a few minutes.
5. Applied DNS, TCP and TLS filters for analysis.

Screenshot 1 – Wireshark Home Screen



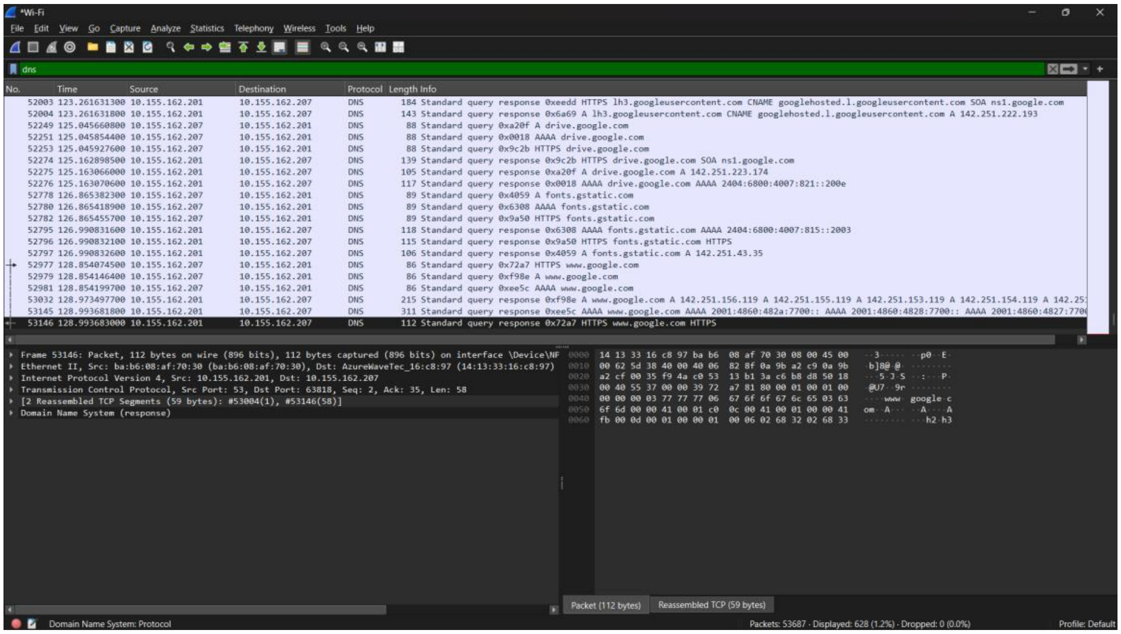
This home screen shows the available network interfaces used for packet capture.

Screenshot 2 – Captured Network Traffic



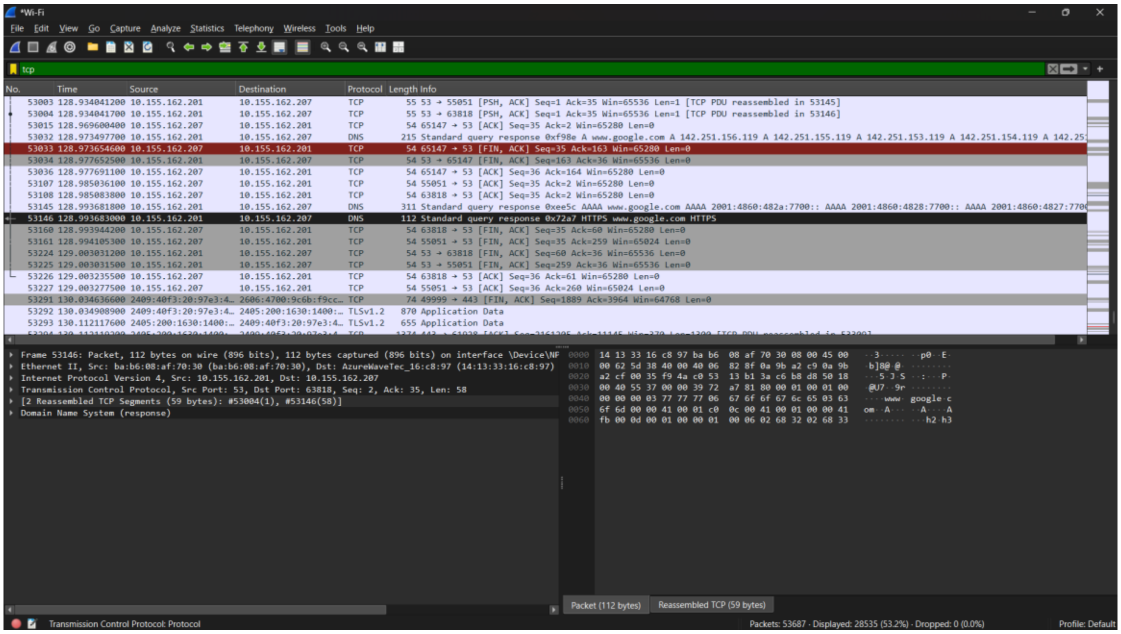
This screenshot shows live packets captured during browsing activity.

Screenshot 3 – DNS Analysis



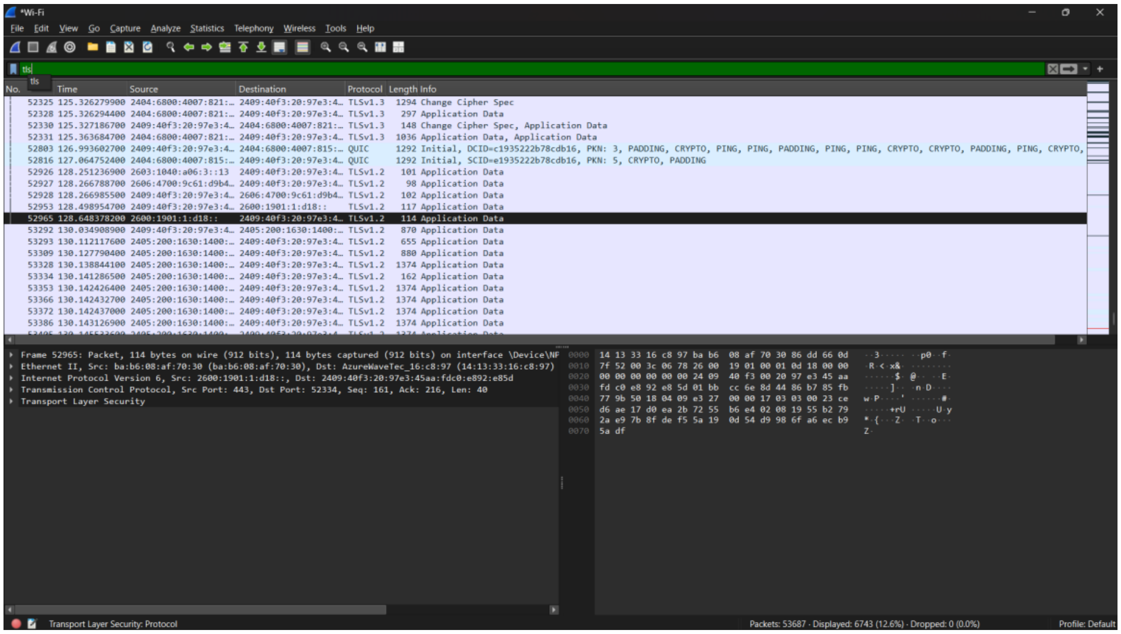
DNS converts domain names into IP addresses. DNS requests and responses were successfully captured.

Screenshot 4 – TCP Analysis



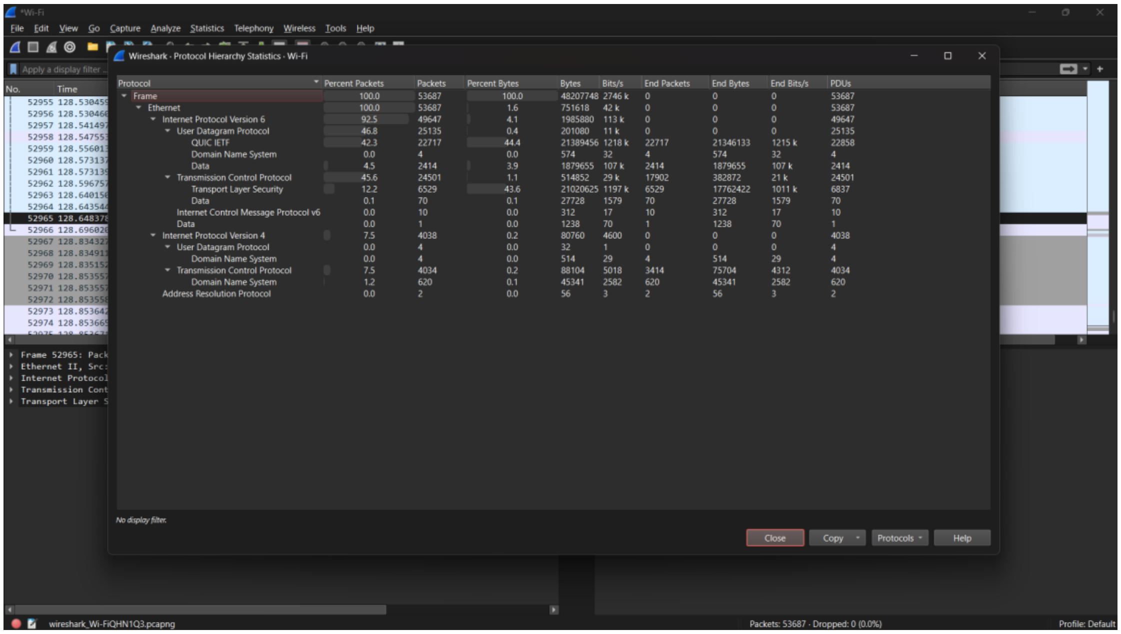
The TCP filter was applied to view TCP packets. Different TCP packets such as ACK and FIN packets were observed, showing communication between the local system and remote servers.

Screenshot 5 – TLS Analysis



During the packet capture process, different types of network traffic were observed. DNS packets were used for domain name resolution. TCP packets were responsible for reliable communication between devices. TLS packets showed encrypted communication with websites.

Screenshot 6 – Protocol Hierarchy Statistics



The protocol hierarchy displays IPv4, IPv6, TCP, UDP, DNS, QUIC and TLS traffic statistics.

ANALYSIS

DNS packets related to Google services were captured. TCP packets showed connection establishment and data transfer. TLS packets confirmed that communication was encrypted. The protocol hierarchy statistics showed that most traffic used IPv6, TCP, QUIC and TLS protocols.

RESULT

The project was successfully completed using Wireshark. Network packets were captured and analyzed using different filters. DNS, TCP, TLS, IPv4 and IPv6 protocols were identified. The captured traffic helped in understanding real-world network communication and security mechanisms..

CONCLUSION

This project helped me understand how data travels through a network and how different protocols are used during communication. By using Wireshark, I gained practical experience in packet capturing and analysis. The project improved my understanding of networking concepts and basic cybersecurity practices.

REFERENCES

1.Wireshark Official Documentation
2. Internship Learning Materials
3. Basic Networking Notes
4. Online Learning Resources